Quiz 7: Hair & DNA Evidence

Review of Session 7. ~15 minutes.

Section A: Hair structure & species

1. Name the three layers of a hair, innermost to outermost.
2. Which cuticle scale pattern belongs to a cat? To a human?
3. Is a human medulla usually continuous or fragmented/absent? An animal medulla?
4. What is hair made of? What pigment gives it color?

Section B: The growth cycle

1. List the four growth phases in order.
2. Which phase are ~85–90% of scalp hairs in, and how long does it last?
3. A hair with a hard club-shaped root and no tissue — was it pulled or shed? What does that mean for DNA?
4. About how fast does human scalp hair grow (per month or per year)?

Section C: Medullary index & the classic questions

1. Write the medullary index formula.
2. Human MI is less than ______. Animal MI is more than ______.
3. Medulla 14 μm, shaft 56 μm. MI? Human or animal?
4. Does hair keep growing after death? Explain.
5. Hair's absorbent property makes it important for detecting what kind of poisoning, and why?

Section D: DNA gel

1. DNA has what charge, so it moves toward which electrode?
2. Do smaller or larger fragments travel farther?
3. How do you find which suspect matches the crime-scene DNA on a gel?

Answer Key

Section A

1. Medulla, cortex, cuticle.
2. Cat = spinous (petal-like); human = imbricate (flat, overlapping).
3. Human = fragmented or absent; animal = continuous (and wide).
4. Keratin; melanin.

Section B

1. Anagen, catagen, telogen, exogen.
2. Anagen; 2–7 years.
3. Shed (telogen); poor for nuclear DNA — only mitochondrial DNA.
4. About 1 cm per month (~½ inch/month, ~6 inches/year).

Section C

1. MI = medulla diameter ÷ hair-shaft diameter.
2. Less than 1/3 (≈0.33); more than 1/2 (≈0.5).
3. 14/56 = 0.25 → human.
4. No — the skin dehydrates and retracts, pulling back from the hair bases so they *appear* longer. No new growth.
5. Arsenic (heavy-metal) poisoning — the metal binds to keratin and stays fixed in the hair, preserving a datable timeline of exposure.

Section D

1. Negative charge → moves toward the positive electrode.
2. Smaller fragments travel farther.
3. Line up the crime-scene lane against each suspect lane; the suspect whose bands sit at the same heights is the match.